NATIONAL UNIVERSITY OF MODERN LANGUAGE ISLAMABAD

Assignment:-01

Submitted to:

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High-Level Design (HLD) for the E-commerce Website:

Frontend Interface:

The frontend interface serves as the user-facing part of the e-commerce website.
It consists of web pages where customers can browse products, view product details, add items to their cart, manage their cart, and complete purchases.

Backend Server:

The backend server handles the business logic, data processing, and communication with external services.

Database:

The database stores essential data for the e-commerce website, including user information, product catalog, inventory status, order details, and transaction history.
It can be implemented using a relational database management system (RDBMS) like MySQL .

User Authentication and Authorization:

This component manages user accounts, authentication, and authorization.
It verifies user credentials during login and generates authentication tokens for secure access to protected resources.
It ensures that only authenticated users can perform certain actions, such as placing orders or accessing account settings.

Product Management:

This component handles the management of the product catalog.
It allows administrators to add, update, or remove products from the catalog, including details such as name, description, price, and availability.

Inventory Management:

This component tracks the availability of products in the inventory.
It updates inventory levels when orders are placed or canceled to prevent overselling or stockouts.

Order Processing:

This component manages the processing of customer orders.
It receives order requests from the frontend interface, verifies product availability, calculates order totals, and generates order confirmation emails.

Payment Processing:

This component handles secure online payments for customer orders.

Low-Level Design (LLD) for the E-commerce Website:

Approaching the low-level design (LLD) of the shopping cart module involves defining the specific components, data structures, and functionalities required to implement this module in detail.

Data Structures:

Define the data structure for the shopping cart, including fields such as product ID, quantity, price, and subtotal.
Decide how the shopping cart data will be stored and managed, either in memory, session storage, or persisted to a database.

API Endpoints:

Define API endpoints for managing the shopping cart, including functionalities such as adding items, removing items, updating quantities, and retrieving cart contents.
Specify the request and response formats for each API endpoint, including error handling and validation.

Business Logic:

Implement business logic for adding items to the cart, ensuring inventory availability, and updating cart totals.
Handle scenarios such as out-of-stock items, quantity limitations, and special offers or discounts.

Concurrency Control:

Implement concurrency control mechanisms to handle simultaneous updates to the shopping cart from multiple users.
Use techniques such as locking, optimistic concurrency control, or versioning to prevent data inconsistencies and race conditions.

Integration with Other Modules:

Integrate the shopping cart module with other backend modules such as the product catalog, inventory management, and order processing.
Ensure proper communication and data exchange between modules to maintain consistency and integrity across the system.

Error Handling and Logging:

Implement error handling mechanisms to handle exceptions, validate input data, and provide meaningful error messages to users.

Security Considerations:

Implement security measures to protect the shopping cart module against threats.
Encrypt sensitive data such as user sessions and payment information to prevent unauthorized.

Testing and Quality Assurance:

Develop unit tests to validate the functionality of the shopping cart module and ensure code correctness.
Perform integration testing to verify interactions with other modules and components of the e-commerce website.
Conduct security testing to identify and address potential vulnerabilities in the shopping cart module.

Documentation:

Document the design decisions, data structures, API specifications, and implementation details of the shopping cart module for future reference and maintenance.